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Roll No

ME-702(A)-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Renewable Energy System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Explain in brief the various types of radiation measuring instruments.
 - b) Discuss with neat sketch a flat plate solar collector. Give a brief insight on the material used for collectors.
2.
 - a) Explain the technology for fabrication of photovoltaic devices. State types of solar cells.
 - b) Discuss Weibull, Rayleigh and normal distribution related to wind data.
3.
 - a) Discuss important elements of WEC systems with neat layout.
 - b) What is capacity factor of wind turbine? How it is helpful in prediction matching the turbine with wind site?

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4.
 - a) Give advantages and disadvantages of small hydro plants.
 - b) Give brief classification of turbines for hydro power plants.
5.
 - a) Explain the constructional details and working of KVIC biogas digester.
 - b) Draw a layout of hydro power plant with function of each component.
6.
 - a) What is wave energy? Explain the working principle of a wave energy conversion system.
 - b) Explain the importance of solar energy power generation with reference to present world energy scenario.
7.
 - a) Discuss any two commercial methods for hydrogen production.
 - b) Define fuel cell. State its working principle. Draw SOFC fuel cell.
8. Write short notes on the following: (any two)
 - a) Solar stills
 - b) Ocean energy conversion
 - c) Selection of hydro turbines
 - d) Origin of Geothermal resources
